Master Learning Plan v6

Senior AI/CV + Life-Sciences-Research Readiness — Master Learning Plan (v6)

Owner: MD Hashim · **Window:** ~mid-Aug 2026 → end-Dec 2027 (~16.5 months) · **Total effort:** ~1,950 focused hours · **Structure:** two blocks + three continuous threads (DSA/C++, implement-by-hand ML, research output) + an application overlay that starts at Month ~3.

> This is v5 kept intact and extended. Nothing from v5 is dropped — every phase, resource, and URL is preserved. v6 adds: a **video** expert layer, a **geometric/molecular-ML** deepening (the differentiator for life-sciences research labs), a light **implement-by-hand ML** muscle-memory thread that runs alongside the DSA thread from Month 3 (so a Research-Scientist pivot stays cheap), a **research-output & networking** thread, **public, conflict-of-interest-safe capstone datasets**, an **application-tier overlay** so you can start applying from Month ~3 and keep learning while interviewing, and a **company target ladder** (relocation-primary, Google-India/India-domestic fallback).

Changelog (v5 → v6) — read this first

Area	v5	v6
Target	Generic senior AI/CV + big-tech	Life-sciences **research** labs (Isomorphic, DeepMind-science) primary; **Google Research India + India medical-imaging AI** kept fully open as fallback
Threads	Thread T: DSA + C++ (from Month 3)	Thread T preserved **+ Thread M: implement-by-hand ML** (same muscle-memory model, light from Month 3, ramps at the crescendo) **+ Thread R: research output & networking**
Video	A few lines inside Phase 8	Full video/temporal expert block + **life-sciences-native video** (DeepLabCut/SLEAP, cell tracking) + a video capstone (Phase 8)
Geometric/molecular ML	Phase 6, moderate	Phase 6 deepened (CS224W, e3nn/NequIP/MACE) + the **Isomorphic-differentiator capstone** on public quantum-chemistry data
Quantum (Phase 4)	~275 hrs	Content & URLs kept; **optional compression to ~180 hrs** noted (you already ship this in production)
Capstones	Private/Lilly data implied	Every capstone re-specified on **public benchmarks**, COI-safe (rules in the new "Conflict-of-interest guardrails" section)
Applications	Deferred to ~Month 12	**Application overlay from Month ~3**; interview while you learn; company ladder added
Publication	"Write 2 posts"	Explicit **paper/preprint/OSS** track (Thread R) + reproduce-a-target-paper habit

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The program at a glance

v6 keeps v5's two-block spine and adds a light continuous scaffold and an early on-ramp to applications.

Block 1 — Depth & foundations (Aug 2026 → ~Mar 2027, ~1,140 hrs). Unchanged in shape: a rapid fundamentals brush-up, then bottom-up mastery — math foundations → your medical-CV domain → DL theory → quantum → production → specialization (geometric/molecular ML). Runs at ~34–38 hrs/week.

Block 2 — Breadth + research/big-tech readiness (~Mar → ~Nov 2027, ~600 hrs + interviewing). GenAI & agentic AI (domain-focused), then a computer-vision *expert* layer (classical/geometric/3D vision + **video** + C++/edge deployment), then a research-and-big-tech interview crescendo (DSA, implement-by-hand ML, system design, research job-talk, behavioral). Runs a touch lighter (~28–32 hrs/week) to leave capacity for interview loops.

Threads (parallel, from ~Month 3). DSA, C++, and implement-by-hand ML are muscle-memory skills — they reward a daily ~40-minute ritual sustained over many months far more than a short cram. So the light threads start once your foundations lock in (~Week 9, early Oct 2026) and run continuously until they feed the Block-2 crescendo. A third weekly thread (research output & networking) runs alongside and carries your live applications.

Application overlay (from ~Month 3.5). You don't wait until the end to interview. Each capstone unlocks an application tier; you enter the funnel at Tier 0 and climb while the plan continues underneath you.

Why this order, and why the threads start at Month 3 (not now)

The smart move for DSA/C++ **and** implement-by-hand ML is Month 3, not today — for two reasons that apply to both. First, Block 1's opening (the Phase 0 brush-up + the Phase 1 math foundations) is the highest-leverage, most focus-hungry stretch of the whole program; diluting it with extra daily disciplines is a bad trade. Second — the point you already intuited for DSA, and which is equally true for implement-by-hand ML — these are **muscle memory**: what builds them is consistency over a long horizon, not front-loaded volume. A ~40-min/day ritual from October to July is ~10 months of spaced repetition, which beats any short intensive block and peaks you right as interviews start.

The implement-by-hand ML thread (Thread M) is deliberately light and starts *after* Phase 1's Karpathy build gives you the substrate (autodiff, attention, a training loop). Keeping it running continuously is precisely what makes a later move toward a **Research Scientist** loop a ramp rather than a restart: research loops ask you to implement ML primitives from a blank file with no libraries, and you'll have been doing exactly that, one primitive a week, for months. Going hardcore up front would cost you the earliest low-hanging fruit (Google-India / India medical-imaging applications), so we keep the *flavour* throughout and save the intensity for the crescendo.

Timeline & weekly load

Block / thread	Window	Weekly load	Content
Block 1 (Phases 0–6)	mid-Aug 2026 → ~Mar 2027	~34–38 hrs/wk	Depth: foundations → CV/segmentation → DL theory → quantum → production → specialization (geometric/molecular)
Thread T (DSA + C++)	early Oct 2026 → ~Oct 2027	+~4 hrs/wk (daily ritual)	Muscle memory: C++ basics → DSA-in-C++, sustained
Thread M (implement-by-hand ML)	early Oct 2026 → ~Oct 2027	+~2 hrs/wk (short ritual)	Muscle memory: reimplement one ML primitive/week from a blank file, unaided
Thread R (research output & networking)	early Oct 2026 → interview season	+~1.5 hrs/wk	Reproduce-a-paper, write, engage labs, run applications
Block 2 (Phases 7–9)	~Mar → ~Nov 2027	~28–32 hrs/wk	Breadth: GenAI/agents → CV expert (+ video) → interview crescendo
Application overlay	~Dec 2026 → end-Dec 2027	inside Thread R	Tiered interviewing while the plan continues

On combined load: during Block 1's second half the three threads push you to ~42–44 hrs/week. That's a lot on top of a full-time role. The threads are deliberately the lightest things on your plate (short daily rituals, often a welcome change of pace) and they are the first things to dial back — to ~2–3 hrs/week total — if any week gets overwhelming, since their value is consistency, not volume. Treat the self-tests and capstones as the real gates, protect one rest block a week, and bank buffer where you can.

Phase summary (the whole plan at a glance)

Block 1 — Depth & foundations · ~1,140 hrs · ~34–38 hrs/wk

Phase	Focus	Effort	Weeks	Approx. dates	Unlocks
0	Rapid brush-up — stats, ML & DL fundamentals	~45 hrs	1–1.5	mid–late Aug 2026	—
1	Foundations — mathematics + neural nets from first principles	~190 hrs	2–6	late Aug → early Oct	—
2	Applied CV & Medical Image Segmentation (+ **Capstone 1**)	~185 hrs	7–11	early Oct → early Nov	**Tier 0**
3	Deep Learning theory & modern architectures	~150 hrs	12–15	Nov → early Dec	**Tier 1**
4	Quantum Computing & QML (compressible to ~180)	~275 hrs	16–22	Dec → late Jan	—
5	Full-stack / production AI systems	~160 hrs	23–26	late Jan → mid-Feb	—
6	Specialization + integration — geometric/molecular ML (+ **Capstone 3**)	~150 hrs	27–30	mid-Feb → mid-Mar 2027	**Tier 2**

Threads T + M + R · ~7.5 hrs/wk combined · Weeks 9 → ~62 (start early Oct 2026)

Block 2 — Breadth + research/big-tech readiness · ~600 hrs + interviewing · ~28–32 hrs/wk

Phase	Focus	Effort	Weeks	Approx. dates	Unlocks
7	GenAI & Agentic AI (domain-focused) (+ **Capstone 4**)	~150 hrs	31–35	mid-Mar → mid-Apr 2027	Tier 2 broadened
8	Computer Vision Expert — classical/geometric/3D **+ video** + C++/edge (+ **Capstone 2** video, **Capstone 5** deploy)	~230 hrs	36–43	mid-Apr → mid-Jun	—
9	Big-Tech + Research Interview Crescendo — DSA, implement-by-hand ML, system design, job-talk, behavioral	~150 hrs	44–48	mid-Jun → mid-Jul	**Tier 3**
—	Active interviewing + maintenance + buffer	—	49–62	mid-Jul → end-Dec 2027	—

Grand total ≈ **1,950 focused hours** across ~16.5 months. Specialized math (optimization, information theory, group theory beyond the Phase 1 core) is threaded into the phase that uses it; its hours are included above.

Threads (the muscle-memory + output scaffold)

All three start ~Week 9 (early Oct 2026) and run continuously. Do them first thing, before the day's phase work, while your head is fresh.

Thread T — DSA + C++ · ~4 hrs/week · Weeks 9 → ~62

(Preserved from v5.) A short daily warm-up, not a block. Durable fluency comes from repetition, so consistency matters more than any one session's length.

Stage T1 — C++ fundamentals (Weeks 9–16, ~32 hrs). Modern C++ so you can then solve problems in it: types, references/pointers, RAII, the STL (vector, map, set, algorithms), classes, templates, move semantics, smart pointers. Serves your hardware-integration need and sets up DSA-in-C++.

- learncpp.com — best free comprehensive C++ course; sections 1–17 — https://www.learncpp.com/
- The Cherno — C++ series (the *why* behind C++ mechanics) — https://www.youtube.com/@TheCherno
- Reference — https://en.cppreference.com/

Stage T2 — DSA in C++, sustained (Weeks 17 → crescendo). ~3–4 problems/week, spaced and revisited, pattern-based not random.

- NeetCode — pattern roadmap + explanations; work the NeetCode 150 — https://neetcode.io/
- LeetCode — the problem bank — https://leetcode.com/
- MIT 6.006 — Introduction to Algorithms — https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/
- Abdul Bari — Algorithms (clearest intuition series) — https://www.youtube.com/@abdul_bari
- Reference text: CLRS, *Introduction to Algorithms*

Thread-T checkpoint (rolling): by end of Block 1, implement from memory in C++ — dynamic arrays, linked lists, stacks/queues, hash maps, binary trees + BST ops, heaps, graph BFS/DFS — and the standard patterns (two-pointer, sliding window, binary search, backtracking, basic DP). Solve a random LeetCode medium in ~30–35 min.

Thread M — Implement-by-hand ML (muscle memory) · ~2 hrs/week · Weeks 9 → ~62 · new in v6

The insurance that keeps a Research-Scientist pivot cheap and sharpens the ML rounds these labs weight most. Each week, from a **blank file, no autocomplete, no AI**, reimplement one primitive in ~30–40 min, then revisit older ones spaced. The Phase 1 Karpathy build is the substrate; this keeps it alive and *timed*.

Rotating primitive list (cycle and revisit): backprop through an MLP · scaled dot-product + multi-head attention · a segmentation loss (Dice/Focal/Tversky) · cross-entropy from logits · LayerNorm / BatchNorm · softmax + numerically-stable log-softmax · a sampler (top-k / nucleus / temperature) · a single diffusion (DDPM) step · logistic regression + its gradient · k-means · PCA via SVD · a tiny SGD/Adam loop · positional encodings (sinusoidal / RoPE) · a KV-cache.

- Dive into Deep Learning (implement-from-scratch textbook, runnable) — https://d2l.ai/
- labml.ai — annotated from-scratch PyTorch implementations — https://nn.labml.ai/
- The Annotated Transformer (line-by-line) — http://nlp.seas.harvard.edu/annotated-transformer/
- Karpathy micrograd / nanoGPT (reference implementations) — https://github.com/karpathy/micrograd · https://github.com/karpathy/nanoGPT
- What the target rounds actually ask (calibrate your drills) — https://igotanoffer.com/en/advice/google-deepmind-research-engineer-interview

Ramp: 1 primitive/week through the phases → implement key components of every capstone from scratch (not just call a library) → in Phase 9 becomes timed, unaided, interviewer-style drills. Because it's continuous, intensifying toward an RS loop later is a gear-change, not new learning.

Thread R — Research output, networking & applications · ~1.5 hrs/week · Weeks 9 → interview season · new in v6

The Tier-2/3 differentiator and your pipeline engine. Rotating weekly:

- **Reproduce-a-target-paper (monthly):** publicly reproduce one paper from a target team (AlphaFold-lineage, DeepMind-science, medical imaging, geometric ML) and write it up. Triple duty: paper-discussion prep, referral hook, proof of level. Find code: https://paperswithcode.com/
- **Write:** 1–2 short posts/month; maintain a "why me for digital biology" positioning narrative.
- **Publish:** aim one capstone at a venue — MICCAI https://www.miccai.org/ · MIDL https://2025.midl.io/ · NeurIPS/ICML/ICLR science workshops incl. **MLSB** (ML for structural biology) https://www.mlsb.io/ and **LoG** https://logconference.org/ · ISMB/RECOMB https://www.iscb.org/ · preprints on arXiv https://arxiv.org/ and bioRxiv https://www.biorxiv.org/ · reviews via OpenReview https://openreview.net/
- **Applications:** run the company ladder (below); from Tier 0 this thread carries your live loops.

BLOCK 1 — Depth & foundations

PHASE 0 — Rapid Brush-Up: Stats, ML & DL Fundamentals

~45 hrs · reactivation for panel readiness · do this first

Goal: rebuild crisp, out-loud command of the fundamentals so opening panel rounds never catch you flat. Reactivation is fast; the aim is fluent articulation, not re-derivation.

Primary resource — StatQuest with Josh Starmer: https://www.youtube.com/@statquest/playlists · index https://statquest.org/

- **0.1 Statistics fundamentals (~12 hrs)** — distributions; mean/variance/SD; populations vs samples; CLT; sampling; hypothesis testing & p-values; confidence intervals; R^2; covariance & correlation; MLE; bootstrapping; power analysis. *Verbal checkpoint:* explain p-value, CI, CLT, MLE each in two sentences, no notes.
- **0.2 Classic ML (~18 hrs)** — bias–variance; train/val/test & CV; confusion matrix, sensitivity/specificity, precision/recall, ROC & AUC (and when PR is honest); linear & logistic regression; ridge (L2) vs lasso (L1) and why L1 → sparsity; trees, random forests, AdaBoost, gradient boosting/XGBoost; SVMs & the kernel trick; k-means & hierarchical clustering; PCA; naive Bayes; entropy & mutual info. *Verbal checkpoint:* cold — gradient boosting vs random forest; why L1 zeros weights and L2 shrinks them (at the gradient level); what AUC measures and when PR-AUC is more honest.
- **0.3 Deep-learning fundamentals (~10 hrs)** — StatQuest Neural Networks playlist + 3Blue1Brown Neural Networks https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi : perceptron/MLP; non-linear activations; backprop ideas; gradient descent; losses; tensors; CNN & RNN/LSTM basics; overfitting, dropout, batch/layer norm, vanishing/exploding gradients; transfer learning.
- **0.4 Deliverable — panel articulation cheat-sheet (~5 hrs)** — one or two crisp spoken sentences per concept. Seeds the war-chest doc.

Phase 0 self-test (verbal, no notes): rapid-fire 15 of the above in 2–3 sentences each, out loud, as if to a panel.

PHASE 1 — Foundations: Mathematics + Neural Networks from First Principles

~190 hrs · the bedrock — do not rush this

Goal: mathematical maturity turned into working neural networks so backprop and "the why" are bone-deep. By the end you can re-derive core results on paper and understand attention mechanically (via Karpathy's GPT build), which unlocks Phase 2 and seeds Thread M.

- **1.1 Linear algebra (~68 hrs) — highest-leverage topic in the plan.**
  - 3B1B Essence of Linear Algebra — https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab
  - MIT 18.06 (Strang) — https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/video_galleries/video-lectures/
  - MIT 18.065 Matrix Methods (Strang) — SVD, PCA, low-rank, GD through a linear-algebra lens — https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/
  - Reference: Matrix Cookbook — https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf
  - *Whiteboard:* derive SVD from eigendecomposition; rank-r approximation geometrically; $A^T A$ is PSD; backprop through a linear layer with matrix calculus.
- **1.2 Calculus + matrix calculus (~22 hrs).**
  - 3B1B Essence of Calculus — https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr
  - Matrix Calculus for Deep Learning (Parr & Howard) — https://explained.ai/matrix-calculus/
- **1.3 Probability & statistics (~45 hrs).**
  - Harvard Stat 110 (Blitzstein) — https://projects.iq.harvard.edu/stat110/youtube
  - Alt: MIT RES.6-012 (Tsitsiklis) — https://ocw.mit.edu/courses/res-6-012-introduction-to-probability-spring-2018/
  - *Whiteboard:* MLE for a Gaussian; conditional expectation as projection; Bayes cleanly on a non-trivial example.
- **1.4 Neural networks from first principles (~55 hrs) — the bridge.** Do all 10 videos with exercises, re-implementing each without looking at his code.
  - Karpathy — Neural Networks: Zero to Hero — https://www.youtube.com/playlist?list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ · hub https://karpathy.ai/zero-to-hero.html (micrograd → makemore 1–5 incl. BatchNorm internals, manual backprop, WaveNet → GPT from scratch → tokenizer → reproduce GPT-2).

Phase 1 self-test: hand-derive backprop for a 2-layer MLP; implement micrograd-style autodiff from memory; explain why BatchNorm helps and what breaks without it; whiteboard scaled dot-product attention and derive the $\sqrt{d_k}$ scaling. **→ Threads T, M, R start now.**

PHASE 2 — Applied Computer Vision & Medical Image Segmentation

~185 hrs · your active domain — PRIORITY: high · ends with Capstone 1 → Tier 0

Goal: own MRI / fluorescence brain segmentation end-to-end. You arrive understanding attention and backprop, so nnU-Net, UNETR, and the loss math land hard.

- **2.1 Imaging physics (~30 hrs).**
  - Allen Elster — Questions & Answers in MRI — https://mriquestions.com/index.html
  - OHBM Educational — MRI physics — https://www.youtube.com/@ohbmonline/playlists
  - Paul Callaghan — "Magnetic Resonance" (Otago), first 4 lectures — https://www.youtube.com/playlist?list=PLqEMVgX2VgZcG_xR1QRdsuc-X_zEgMx-Z
  - iBiology — fluorescence microscopy primer — https://www.youtube.com/playlist?list=PLF513KEDjY9YBNhwMfX6jMl3PlpdW9TMP
  - *Deliverable:* 1-page note — T1/T2/FLAIR/T1CE + fluorescence artifacts (photobleaching, bleed-through, autofluorescence).
- **2.2 CNNs + convolution math + U-Net family + losses (~50 hrs).**
  - CS231n 2017 (Johnson) — https://www.youtube.com/playlist?list=PL3FW7Lu3i5JvHM8ljYj-zLfQRF3EO8sYv · https://cs231n.github.io/
  - Convolution arithmetic (Dumoulin & Visin) — https://arxiv.org/abs/1603.07285 · https://github.com/vdumoulin/conv_arithmetic
  - U-Net https://arxiv.org/abs/1505.04597 · V-Net https://arxiv.org/abs/1606.04797 · Attention U-Net https://arxiv.org/abs/1804.03999 · nnU-Net https://www.nature.com/articles/s41592-020-01008-z
  - Losses (your 95/5 imbalance) — survey https://arxiv.org/abs/2006.14822 · Generalized Dice https://arxiv.org/abs/1707.03237 · Focal Tversky https://arxiv.org/abs/1810.07842 · Boundary loss https://arxiv.org/abs/1812.07032 (implement Dice/GDL/Focal/Tversky/Boundary/compound from scratch — Thread M synergy).
  - *Whiteboard:* draw U-Net with tensor shapes; why skip connections work; Dice gradient instability on empty masks; nnU-Net's "no new architecture" thesis.
- **2.3 nnU-Net + MONAI + microscopy-native hands-on (~45 hrs) — biggest applied block.**
  - MONAI tutorials — https://github.com/Project-MONAI/tutorials · videos https://www.youtube.com/@projectmonai/videos
  - nnU-Net v2 source — https://github.com/MIC-DKFZ/nnUNet · MONAI integration https://github.com/Project-MONAI/tutorials/blob/main/nnunet/README.md · Kitware comparison https://www.kitware.com/developing-custom-3d-medical-image-segmentation-solutions-using-out-of-the-box-pipelines-in-monai/
  - DigitalSreeni (Python for Microscopists) — https://www.youtube.com/@DigitalSreeni/playlists · code https://github.com/bnsreenu/python_for_microscopists · U-Net playlist https://www.youtube.com/playlist?list=PLZsOBAyNTZwbR08R959iCvYT3qzhxvGOE
  - AI for Medical Diagnosis (Rajpurkar) — audit free — https://www.coursera.org/learn/ai-for-medical-diagnosis
- **2.4 Transformer-based segmentation + foundation models (~20 hrs).**
  - ViT https://arxiv.org/abs/2010.11929 · TransUNet https://arxiv.org/abs/2102.04306 · Swin-UNet https://arxiv.org/abs/2105.05537 · UNETR https://arxiv.org/abs/2103.10504 · Swin UNETR https://arxiv.org/abs/2201.01266
  - SAM https://arxiv.org/abs/2304.02643 · MedSAM https://www.nature.com/articles/s41467-024-44824-z · MedNeXt https://arxiv.org/abs/2303.09975
- **2.5 3D, uncertainty, evaluation (~40 hrs).**
  - TorchIO — https://torchio.readthedocs.io/ · https://www.youtube.com/playlist?list=PLfXIfDgnRPlfIEHFx2N6mlbn-MGFOPiX2
  - Uncertainty (Yarin Gal — MC dropout, ensembles, evidential) — https://www.youtube.com/results?search_query=yarin+gal+uncertainty+deep+learning
  - Metrics Reloaded — https://www.nature.com/articles/s41592-023-02151-z
  - Domain shift (Stanford AIMI) — https://aimi.stanford.edu/education/educational-resources

- *3D-conv exercise:* input (D=64,H=128,W=128,C=4), kernel (3,3,3), stride (1,2,2), padding (1,1,1), 32 out-channels — compute output shape + param count by hand; repeat for transposed + dilated.

PHASE 2 — CAPSTONE 1 (~15 hrs) · public data, COI-safe · unlocks Tier 0. Notebook + 3-page write-up: (1) preprocessing with physics justification, (2) architecture ablation across 3 architectures, (3) loss-function ablation, (4) uncertainty + failure-mode analysis. Implement the losses and one architecture block from scratch (Thread M).

- **Datasets (pick one):** Medical Segmentation Decathlon (Task01 BrainTumour = BraTS subset, CC-BY-SA) http://medicaldecathlon.com/ · BraTS https://www.synapse.org/brats · The Cancer Imaging Archive https://www.cancerimagingarchive.net/ · IXI (open brain MRI) https://brain-development.org/ixi-dataset/ — **or microscopy-native:** LIVECell https://github.com/sartorius-research/LIVECell · Sartorius Cell Instance Segmentation (Kaggle) https://www.kaggle.com/competitions/sartorius-cell-instance-segmentation · Broad Bioimage Benchmark Collection https://bbbc.broadinstitute.org/
- **Angle:** portfolio + a MIDL/MICCAI-workshop-grade short paper (Thread R). Directly Qure.ai / SigTuple / Google-India relevant.

Phase 2 self-test: whiteboard U-Net forward pass + skips with shapes; why nnU-Net usually beats fancier architectures (and when not); diagnose bias-field artifacts, class imbalance, label noise, domain shift on a new dataset.

PHASE 3 — Deep Learning Theory & Modern Architectures

~150 hrs · the broad, deep DL backbone · unlocks Tier 1

Goal: generalize beyond Phase 1–2 — advanced transformer variants, self-supervised learning, generative/diffusion theory, training-dynamics math.

- **3.1 Optimization for deep learning (~28 hrs).**
  - Sebastian Ruder — https://www.ruder.io/optimizing-gradient-descent/
  - Stanford EE364A Convex Optimization (Boyd) — https://www.youtube.com/playlist?list=PL3940DD956CDF0622 · book https://web.stanford.edu/~boyd/cvxbook/
  - Optimizer/training-dynamics deep dive (SGD, momentum, Adam, warmup, schedules, second-order intuitions).
- **3.2 Information theory for ML (~18 hrs).**
  - David MacKay — Information Theory, Inference, and Learning Algorithms, ch 1–6 — https://www.inference.org.uk/itila/book.html
  - *Whiteboard:* cross-entropy from KL; NLL minimization = MLE; the ELBO; mutual information in InfoNCE.
- **3.3 Transformers in depth + modern variants (~40 hrs).**
  - Stanford CS25 — Transformers United — https://web.stanford.edu/class/cs25/ · https://www.youtube.com/playlist?list=PLoROMvodv4rNiJRchCzutFw5ItR_Z27CM
  - Lilian Weng — https://lilianweng.github.io/
  - The Annotated Transformer — http://nlp.seas.harvard.edu/annotated-transformer/
  - RoPE https://arxiv.org/abs/2104.09864 · RMSNorm https://arxiv.org/abs/1910.07467 · FlashAttention https://arxiv.org/abs/2205.14135 · GQA/MQA https://arxiv.org/abs/2305.13245 · SwiGLU https://arxiv.org/abs/2002.05202 · MoE/Switch https://arxiv.org/abs/2101.03961
- **3.4 Vision transformers + modern ConvNets (~25 hrs).**
  - ViT lectures (Lucas Beyer) — https://www.youtube.com/results?search_query=lucas+beyer+vision+transformer
  - ConvNeXt https://arxiv.org/abs/2201.03545 · DINOv2 https://arxiv.org/abs/2304.07193 · MAE https://arxiv.org/abs/2111.06377
  - Hugging Face Computer Vision course — https://huggingface.co/learn/computer-vision-course
- **3.5 Generative modeling & diffusion (~38 hrs).**
  - Stanford CS236 (Ermon) — https://www.youtube.com/playlist?list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8
  - DDPM https://arxiv.org/abs/2006.11239 · Yang Song score-based blog https://yang-song.net/blog/2021/score/ · Lilian Weng diffusion https://lilianweng.github.io/posts/2021-07-11-diffusion-models/
  - Hugging Face Diffusion course (units 1–2) — https://huggingface.co/learn/diffusion-course

Phase 3 self-test: derive the diffusion loss (variational + score-matching views); compare positional encodings; explain self-supervised pretraining in low-data regimes. **→ Tier 1 (broader CV/DL roles).**

PHASE 4 — Quantum Computing & Quantum Machine Learning

~275 hrs (optionally compress to ~180) · rigorous theory beneath your existing hands-on work

Goal: match your production VQE/ADAPT-VQE/DMET-VQE experience with real theory. Derive VQE from the variational principle, explain barren plateaus mathematically, defend ansatz choices, and discuss QML's genuine limits honestly (Schuld).

> **v6 tweak — optional compression to ~180 hrs.** You already ship this in production and the relocation targets (Isomorphic/DeepMind-science) are DL/GNN/generative, not quantum computing. If time-constrained: keep 4.1–4.2 (as needed), 4.4 (VQA math + barren plateaus), 4.5 (Schuld critique), and Capstone; treat 4.3 as consolidation and **defer** the deep error-correction course and QDA breadth unless you target a quantum-specific role. All content and URLs below are kept regardless.

- **4.1 Group theory & symmetry (~22 hrs).**
  - Group theory for physicists — https://www.youtube.com/results?search_query=group+theory+for+physicists+lectures
  - Geometric Deep Learning lectures (Bronstein) — the Lie-group/equivariance angle — https://geometricdeeplearning.com/lectures/
- **4.2 QM math: Hilbert spaces, operators, tensor products (~25 hrs).**
  - MIT 8.04 (Adams) — https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/video_galleries/lecture-videos/
  - Quantum Country — https://quantum.country/qcvc
  - (Optional) Schuller — Geometric Anatomy of Theoretical Physics — https://www.youtube.com/playlist?list=PLPH7f_7ZlzxTi6kS4vCmv4ZKm9u8g5yic

- **4.3 Quantum computing fundamentals — Watrous / IBM track (~80 hrs).**
  - Basics of Quantum Information — https://learning.quantum.ibm.com/course/basics-of-quantum-information · YouTube https://www.youtube.com/playlist?list=PLOFEBzvs-VvrXTMy5Y2IqmSaUjfnhvBHR
  - Fundamentals of Quantum Algorithms — https://learning.quantum.ibm.com/course/fundamentals-of-quantum-algorithms
  - General Formulation of Quantum Information — https://learning.quantum.ibm.com/course/general-formulation-of-quantum-information
  - Foundations of Quantum Error Correction — https://learning.quantum.ibm.com/course/foundations-of-quantum-error-correction
  - Companion: Nielsen & Chuang, ch 1–4, 8, 10
- **4.4 Variational quantum algorithms — VQE, ADAPT-VQE, DMET, QAOA (~45 hrs).**
  - PennyLane Codebook — https://pennylane.ai/codebook
  - PennyLane quantum-chemistry demos — https://pennylane.ai/qml/demos_quantum-chemistry
  - VQE https://www.nature.com/articles/ncomms5213 · ADAPT-VQE https://www.nature.com/articles/s41467-019-10988-2 · DMET-VQE https://arxiv.org/abs/2108.08987 · QAOA https://arxiv.org/abs/1411.4028 · Hadfield review https://arxiv.org/abs/1709.03489 · Pauli grouping https://arxiv.org/abs/1907.13117
- **4.5 Quantum machine learning theory (~42 hrs).**
  - PennyLane QML topic (Schuld-curated) — https://pennylane.ai/topics/quantum-machine-learning
  - Maria Schuld — "Taking stock of QML: a critical perspective" — https://www.youtube.com/results?search_query=maria+schuld+taking+stock+quantum+machine+learning
  - Schuld & Petruccione, *Machine Learning with Quantum Computers*, ch 4–7
  - QML=kernels https://arxiv.org/abs/2101.11020 · Barren plateaus https://www.nature.com/articles/s41467-018-07090-4 · Expressibility https://arxiv.org/abs/1905.10876 · Advantage-goal https://arxiv.org/abs/2203.01340 · PQC encodings https://arxiv.org/abs/2008.08605
  - Courses to pull from: Péré https://github.com/Christophe-pere/QML-Course · Hjorth-Jensen https://github.com/CompPhysics/QuantumComputingMachineLearning
- **4.6 Quantum chemistry on quantum computers + capstone (~60 hrs).**
  - Qiskit Global Summer School — https://www.youtube.com/@qiskit/playlists
  - IBM Quantum Learning + Sample-based/Krylov Quantum Diagonalization — https://learning.quantum.ibm.com/ · https://www.ibm.com/quantum/blog/iql-migration

PHASE 4 — CAPSTONE 3b (~30 hrs) · public molecules, COI-safe. Take a **public** molecule (textbook H_2 / LiH / H_2O, or a QM9 subset — **not** your Lilly molecules such as the keto-enol/HATU work). Re-implement three ways — vanilla VQE + UCCSD, ADAPT-VQE, DMET-VQE — benchmark against classical CCSD; 4-page write-up defends every choice (ansatz, optimizer, measurement grouping, noise mitigation, basis set) and includes a barren-plateau analysis. Optionally add a small classical-ML surrogate (delta-ML toward CCSD) to make it a "quantum × ML" story.

Phase 4 self-test: derive VQE from the variational principle; parameter-shift rule and why it works; barren-plateau argument; VQE vs QAOA — when each and when no quantum algorithm; honest take on QML hype vs promise (cite Schuld); DMET fragmentation and the classical-quantum interface.

PHASE 5 — Full-Stack / Production AI Systems

~160 hrs · production breadth + systems-design readiness

Goal: upgrade from "I built it" to "I can defend every architectural choice and design the next-gen version" across LLM systems, RAG, MLOps, distributed training, serving.

- **5.1 LLM internals (~32 hrs).**
  - Stanford CS336 — Language Modeling from Scratch — https://stanford-cs336.github.io/spring2024/ · https://www.youtube.com/playlist?list=PLoROMvodv4rOY23Y0BoGoBGgQ1zmU_MT_
  - Hugging Face NLP course — https://huggingface.co/learn/nlp-course
- **5.2 RAG at senior level (~28 hrs).**
  - Hamel Husain — "What we learned from a year of building with LLMs" — https://www.oreilly.com/radar/what-we-learned-from-a-year-of-building-with-llms-part-i/
  - Eugene Yan — LLM patterns — https://eugeneyan.com/writing/llm-patterns/
  - Anthropic — contextual retrieval — https://www.anthropic.com/news/contextual-retrieval
  - LlamaIndex https://docs.llamaindex.ai/ · RAGAS https://docs.ragas.io/ · DSPy https://dspy.ai/
- **5.3 MLOps + ML systems design (~42 hrs).**
  - Chip Huyen — ML systems design / interviews book — https://huyenchip.com/ml-interviews-book/contents/8.1.1-ml-systems-design.html
  - Stanford CS329S — https://stanford-cs329s.github.io/
  - Made With ML — https://madewithml.com/ · Full Stack Deep Learning — https://fullstackdeeplearning.com/course/2022/
- **5.4 Distributed training + inference optimization (~25 hrs).**
  - Sebastian Raschka — https://magazine.sebastianraschka.com/
  - HF — efficient training on multiple GPUs — https://huggingface.co/docs/transformers/en/perf_train_gpu_many
  - vLLM https://arxiv.org/abs/2309.06180 · quantization (GPTQ/AWQ/GGUF/FP8) · ZeRO https://arxiv.org/abs/1910.02054
- **5.5 Eval / observability / safety + systems-design prep (~33 hrs).**
  - Anthropic — Constitutional AI — https://www.anthropic.com/news/constitutional-ai · Eugene Yan — LLM evals — https://eugeneyan.com/writing/llm-evaluators/
  - Observability (LangSmith / Phoenix / Helicone) — run one on a real RAG system.
  - Systems-design prep — Chip Huyen's book https://huyenchip.com/ml-interviews-book/ ; whiteboard, recorded, self-critiqued: "RAG assistant for HCPs, 10K queries/day on 50K docs"; "brain-seg pipeline ingesting new-institution data weekly with domain-shift handling"; "retraining + monitoring for a drug-discovery digital twin."

PHASE 6 — Specialization + Integration (Geometric / Molecular ML)

~150 hrs · the culmination — ties imaging, quantum, and drug-discovery together · ends with Capstone 3 → Tier 2

Goal: close the geometric-deep-learning / molecular-AI gap that connects all your work — **the single highest-reward phase for the relocation target** (Isomorphic/DeepMind-science live in this lane, and your quantum-chem background makes it uniquely credible).

- **6.1 Geometric deep learning + graph ML (~50 hrs).**
  - **Stanford CS224W — Machine Learning with Graphs (Leskovec)** — the canonical GNN course — https://web.stanford.edu/class/cs224w/ · 2023 https://snap.stanford.edu/class/cs224w-2023/ · playlist https://www.youtube.com/playlist?list=PLoROMvodv4rPLKxIpqhjhPgdQy7imNkDn *(new in v6)*
  - Bronstein et al. — Geometric Deep Learning — https://geometricdeeplearning.com/lectures/ · proto-book https://arxiv.org/abs/2104.13478 · Veličković GNN lectures https://www.youtube.com/results?search_query=petar+velickovic+gnn+lectures
  - Libraries: PyTorch Geometric https://pytorch-geometric.readthedocs.io/ · DGL https://www.dgl.ai/ *(new in v6)*
- **6.2 Molecular ML + equivariant networks (~50 hrs).**
  - DeepChem https://deepchem.io/ · tutorials https://deepchem.io/tutorials/ · Therapeutics Data Commons https://tdcommons.ai/
  - SchNet https://arxiv.org/abs/1706.08566 · DimeNet++ https://arxiv.org/abs/2011.14115 · GemNet https://arxiv.org/abs/2106.08903 · MolFormer https://arxiv.org/abs/2106.09553 · equivariant nets (Cohen & Welling) https://arxiv.org/abs/1602.07576
  - **Equivariant NN stack (new in v6):** EGNN https://arxiv.org/abs/2102.09844 · e3nn https://e3nn.org/ · https://github.com/e3nn/e3nn · tutorial https://blondegeek.github.io/e3nn_tutorial/ · NequIP https://github.com/mir-group/nequip · paper https://www.nature.com/articles/s41467-022-29939-5 · MACE https://github.com/ACEsuit/mace · paper https://arxiv.org/abs/2206.07697 · Allegro https://github.com/mir-group/allegro · Equiformer https://arxiv.org/abs/2206.11990 · curated list https://github.com/Eipgen/Neural-Network-Models-for-Chemistry
- **6.3 Protein structure + equivariance (~25 hrs).**
  - AlphaFold 2 — AlQuraishi explainer https://www.youtube.com/results?search_query=AlphaFold+2+explained+AlQuraishi · paper https://www.nature.com/articles/s41586-021-03819-2
  - ESM/ESMFold https://github.com/facebookresearch/esm · ESM Atlas https://esmatlas.com/ · **Boltz (open structure model) https://github.com/jwohlwend/boltz** · OpenFold https://github.com/aqlaboratory/openfold · e3nn/EGNN *(links above) (new in v6)*
- **6.4 Mock interviews + portfolio + writing (~25 hrs).** Recorded, reviewed: 45-min segmentation deep-dive; 45-min quantum (defend VQE/ADAPT/DMET to a skeptical chemist); 30-min behavioral (STAR, 4 stories). Read 1 paper/day; write 2 posts (Thread R).

PHASE 6 — CAPSTONE 3 (~included above) · public quantum-chemistry data, COI-safe · unlocks Tier 2 · the Isomorphic-differentiator piece. Train an **E(3)-equivariant GNN** (SchNet → DimeNet++ → EGNN/NequIP-style) to predict molecular properties; ablate equivariance; discuss the DFT-vs-quantum-computing data-generation angle (ties to your VQE work). Implement the message-passing core from scratch (Thread M).

- **Datasets (public):** QM9 https://quantum-machine.org/datasets/ · **OGB-LSC PCQM4Mv2 (HOMO-LUMO gap, 3.7M molecules)** https://ogb.stanford.edu/kddcup2021/pcqm4m/ · MoleculeNet https://moleculenet.org/ · OGB https://ogb.stanford.edu/ · TDC https://tdcommons.ai/
- **Angle:** portfolio + an MLSB / AI4Science / LoG workshop paper (Thread R). The piece that says "I can do what Isomorphic does," on 100% public benchmarks.

Phase 6 self-test: message passing vs convolution; what "equivariance" buys you and why it matters for molecules; when a GNN beats a fingerprint baseline; how AlphaFold's structure module uses equivariance.

BLOCK 2 — Breadth, GenAI & Big-Tech/Research Readiness

~Mar → ~Nov 2027 · ~600 hrs + interviewing · ~28–32 hrs/week · Threads T + M + R continue underneath

PHASE 7 — Generative AI & Agentic AI (domain-focused)

~150 hrs · Weeks 31-35 (~mid-Mar → mid-Apr 2027) · ends with Capstone 4

Goal: move from "I use LLMs in a RAG app" to designing, adapting, and orchestrating modern GenAI/agentic systems — at a *competent, domain-focused* level (protein/molecular LMs, multimodal medical VLMs, literature agents), not generic chatbot RAG.

- **7.1 Advanced LLMs — adaptation & inference (~40 hrs).**
  - LoRA https://arxiv.org/abs/2106.09685 · QLoRA https://arxiv.org/abs/2305.14314 · Raschka's practical LoRA https://magazine.sebastianraschka.com/
  - InstructGPT/RLHF https://arxiv.org/abs/2203.02155 · DPO https://arxiv.org/abs/2305.18290 · Hugging Face TRL
  - Hugging Face LLM course — https://huggingface.co/learn/llm-course
  - Chip Huyen — *AI Engineering* (O'Reilly, 2025) — read alongside.
- **7.2 Multimodal & vision-language models (~25 hrs) — also feeds Phase 8.**
  - CLIP https://arxiv.org/abs/2103.00020 · LLaVA https://arxiv.org/abs/2304.08485 · Flamingo https://arxiv.org/abs/2204.14198
- **7.3 Agents & agentic systems (~50 hrs) — the core.**
  - Hugging Face Agents Course (certified) — https://huggingface.co/learn/agents-course
  - Hugging Face MCP Course — https://huggingface.co/learn/mcp-course
  - Anthropic — Building Effective Agents — https://www.anthropic.com/research/building-effective-agents
  - Andrew Ng — agentic design patterns (DeepLearning.AI short courses) — https://www.deeplearning.ai/short-courses/
  - LangGraph — https://langchain-ai.github.io/langgraph/
  - Patterns: ReAct, tool/function calling, reflection/self-critique, planning/decomposition, memory, multi-agent; plus agent evaluation.

PHASE 7 — CAPSTONE 4 (~35 hrs) · public literature, COI-safe. A multi-step research/analysis agent (RAG + tool use + memory + eval harness) over **open** scientific literature (medical imaging or drug discovery): "given a target/modality, retrieve → synthesize → self-critique recent methods." Write up the failure modes you engineered around.

- **Corpora/tools (public):** PMC Open Access https://www.ncbi.nlm.nih.gov/pmc/tools/openftlist/ · arXiv https://arxiv.org/ · PubMed E-utilities http s://www.ncbi.nlm.nih.gov/books/NBK25501/ · Semantic Scholar API https://www.semanticscholar.org/product/api · structured tool-calling via TDC/ChEMBL. **Never ingest Lilly internal docs/data.**

Phase 7 self-test: explain LoRA and why it's parameter-efficient; RLHF vs DPO; when an agent beats a single well-prompted call (and when it's needless complexity); walk the ReAct loop; how you'd evaluate an agent's reliability. **→ Tier 2 broadened.**

PHASE 8 — Computer Vision Expert (+ Video + C++/edge)

~230 hrs · Weeks 36–43 (~mid-Apr → mid-Jun 2027) · ends with Capstone 2 (video) + Capstone 5 (deploy)

Goal: round out into a broad CV expert — classical & geometric vision, 3D, **video/temporal**, detection/tracking — plus the deployment/optimization skills that put CV models onto real hardware (your integration need). Block 1's Phases 2–3 are the modern-DL half; this is the classical/geometric half **plus the video half plus the systems half**, and it's where the C++ thread pays off.

- **8.1 Classical & geometric computer vision (~45 hrs) — the foundation most ML engineers lack.**
  - First Principles of Computer Vision — Shree Nayar (Columbia) — https://fpcv.cs.columbia.edu/ · Coursera specialization https://www.coursera.org/specializations/firstprinciplesofcomputervision
  - Cyrill Stachniss — Photogrammetry & Mobile Sensing (multi-view geometry, SLAM, point clouds) — https://www.youtube.com/@CyrillStachniss
  - Own: camera models & calibration, homography, epipolar geometry, stereo, SIFT/ORB, optical flow, SfM, visual SLAM.
  - Reference: Hartley & Zisserman, *Multiple View Geometry in Computer Vision*
- **8.2 Modern DL for vision — breadth (~35 hrs).**
  - Michigan EECS 498 — Deep Learning for Computer Vision (Justin Johnson) — https://web.eecs.umich.edu/~justincj/teaching/eecs498/ · playlist https://www.youtube.com/playlist?list=PL5-TkQAfAZFbzxjBHtzdVCWE0Zbhomg7r
  - Detection (two-stage vs one-stage), instance/panoptic segmentation, pose estimation.
- **8.3 Video & temporal analytics (~40 hrs) — new in v6.**
  - **Video transformers / action recognition:** VideoMAE https://arxiv.org/abs/2203.12602 · TimeSformer https://arxiv.org/abs/2102.05095 · ViViT https://arxiv.org/abs/2103.15691 · SlowFast https://arxiv.org/abs/1812.03982
  - **Promptable video segmentation:** SAM 2 https://arxiv.org/abs/2408.00714 · https://github.com/facebookresearch/sam2 · https://ai.meta.com/sam2 (extends your SAM/MedSAM work)
  - **Optical flow:** RAFT https://arxiv.org/abs/2003.12039 · https://github.com/princeton-vl/RAFT
  - **Tracking (beyond DeepSORT):** ByteTrack https://arxiv.org/abs/2110.06864 · https://github.com/ifzhang/ByteTrack · OC-SORT https://arxiv.org/abs/2203.14360 · https://github.com/noahcao/OC_SORT · BoT-SORT https://arxiv.org/abs/2206.14651 · https://github.com/NirAharon/BoT-SORT
  - **Metrics & benchmarks:** HOTA https://arxiv.org/abs/2009.07736 · MOTChallenge https://motchallenge.net/ · VOS datasets DAVIS https://davischallenge.org/ · YouTube-VOS https://youtube-vos.org/
- **8.4 Life-sciences-native video (~30 hrs) — the differentiator, new in v6.** Where video analytics meets your domain and becomes a portfolio piece target teams recognize.
  - **Animal-behavior pose & tracking:** DeepLabCut https://www.deeplabcut.org/ · https://github.com/DeepLabCut/DeepLabCut · Nature Neuroscience https://www.nature.com/articles/s41593-018-0209-y · SLEAP https://sleap.ai/ · https://github.com/talmolab/sleap · Nature Methods https://www.nature.com/articles/s41592-022-01426-1
  - **Live-cell microscopy time-series:** Cellpose https://www.cellpose.org/ · https://github.com/MouseLand/cellpose · TrackMate https://imagej.net/plugins/trackmate/ · Ultrack https://github.com/royerlab/ultrack · Cell Tracking Challenge http://celltrackingchallenge.net/
- **8.5 Deployment & optimization for hardware/edge (~50 hrs) — CV expert meets integration.**
  - OpenCV in C++ — https://docs.opencv.org/ · applied recipes https://pyimagesearch.com/
  - On-device serving: ONNX https://onnx.ai/ · NVIDIA TensorRT https://developer.nvidia.com/tensorrt · quantization & pruning for CV models.
  - CUDA basics — enough to reason about GPU kernels and data movement (NVIDIA intro materials).
- **8.6 3D vision & neural rendering (~30 hrs).**
  - NeRF https://arxiv.org/abs/2003.08934 · 3D Gaussian Splatting https://arxiv.org/abs/2308.04079 · depth estimation, point-cloud processing.

PHASE 8 — CAPSTONE 2 (video, ~included) · public data, COI-safe · unlocks/strengthens Tier 1. DeepLabCut/SLEAP pose → temporal model → behavior classification; **or** Cellpose seg → TrackMate/Ultrack tracking + lineage; use SAM 2 for mask propagation.

- **Datasets (public):** CalMS21 (Caltech mouse social behavior) https://data.caltech.edu/records/s0vdx-0k302 · MABe challenge https://www.aicrowd.com/challenges/multi-agent-behavior-challenge-2022 · Cell Tracking Challenge http://celltrackingchallenge.net/ · (general variant) DAVIS/YouTube-VOS above.

PHASE 8 — CAPSTONE 5 (deployment, ~included) · public data, COI-safe. A geometric-CV or 3D task deployed with a C++/optimized inference component: camera calibration + SfM on public sets, or feature-based visual odometry, or take C1/C2's model and get it running optimized (TensorRT/ONNX, quantized) behind a C++ path.

- **Datasets (public):** KITTI https://www.cvlibs.net/datasets/kitti/ · TUM RGB-D https://cvg.cit.tum.de/data/datasets/rgbd-dataset · ETH3D https://www.eth3d.net/

Phase 8 self-test: derive the pinhole camera model and the epipolar constraint; SIFT and why scale/rotation invariance matters; two-stage vs one-stage detectors; SAM 2's memory mechanism for mask propagation; tracking-by-detection vs joint detection-and-tracking; what TensorRT does to speed up inference; how you'd port a PyTorch CV model to a real-time C++ pipeline.

PHASE 9 — Big-Tech + Research Interview Crescendo

~150 hrs · Weeks 44–48 (~mid-Jun → mid-Jul 2027), then maintenance through interview season · unlocks Tier 3

Goal: peak the interview-specific skills right as you interview at Google-caliber and research-lab targets. This is where the year-long threads convert into performance — you're sharpening reflexes you already have, not starting cold. **Recalibrated for research tracks, not a generic**

SWE loop.

- **9.1 DSA — interview-grade (~55 hrs) — the crescendo of Thread T.** Timed NeetCode 150 → Blind 75 → company-tagged sets — https://neetcode.io/ · https://leetcode.com/ ; pattern mastery under pressure — DesignGurus, Grokking the Coding Interview https://www.designgurus.io/ . Do problems out loud, blank editor, ~35 min, **code must run** (practice CoderPad-style — a DeepMind quirk); ~4–6/day this phase.
- **9.2 Implement-by-hand ML (~35 hrs) — the crescendo of Thread M, new in v6.** Timed, unaided, from a blank file: attention, a loss, a sampler, a diffusion step, a small training loop, backprop. *This is literally a DeepMind RE ML-round task, and where an RS pivot pays off.* Plus an **ML math/theory** sweep: gradient computation, KL properties, sampling, convergence intuition, the "L1 sparsity via gradients" mechanism answer (asked verbatim).
- **9.3 System design (~30 hrs) — distinct from Phase 5's ML-systems-design.** ByteByteGo / Alex Xu — https://bytebytego.com/ · https://www.youtube.com/@ByteByteGo ; Designing Data-Intensive Applications (Kleppmann) — book ; Gaurav Sen — https://www.youtube.com/@gkcs ; optional MIT 6.824 — https://pdos.csail.mit.edu/6.824/ . Whiteboard 8–10 classic + ML-flavored prompts, recorded, self-critiqued.
- **9.4 Research job-talk + behavioral (~30 hrs) — new/expanded in v6.** A 20-min talk + Q&A on each of your two strongest capstones (expect constraints piled on iteratively). Structure ~6–8 STAR stories mapped to each org's signals; low-level/OO design (parking lot, elevator, rate limiter) for loops that include it; live peer mocks — Pramp / interviewing.io. **Research loops ban AI tools — rehearse fully unaided.**

Phase 9 self-test: a random LeetCode hard in ~45 min with clean, running code; implement attention + a loss + a sampler from a blank file, unaided, timed; a full system-design whiteboard in 45 min with capacity estimates and tradeoffs; a crisp 20-min job-talk on a capstone; four STAR stories from memory. **→ Tier 3 (top research loops).**

(Weeks 49–62, ~mid-Jul → end-Dec 2027: active interviewing. Keep DSA, implement-by-hand ML, and system design warm with ~5 hrs/week maintenance; use the buffer for on-sites, take-homes, and rest.)

Application overlay & company target ladder (new in v6)

You enter the funnel at Tier 0 and climb while the plan continues. Match the tier; treat early loops as calibration you get paid in feedback for. "Relocate" flags roles outside India.

Tier	Unlocked after	~When	Apply to
Tier 0	Capstone 1 (segmentation)	~early Nov 2026	India medical-imaging AI + accessible; stretch Google-India
Tier 1	Capstone 2/3 (video / theory)	~Dec 2026 → Jun 2027	broader CV/video + biomedical-imaging; stronger Google-India case
Tier 2	Capstone 3 (molecular GNN)	~mid-Mar 2027 →	AI drug-discovery labs; **Isomorphic / DeepMind-science RE realistic**
Tier 3	Interview crescendo (P9)	~Jul 2027 →	full top-lab loops

Tier 0 — India-domestic + accessible: Qure.ai (Mumbai/Bengaluru — radiology imaging) https://qure.ai/ · SigTuple (Bengaluru — microscopy/cell; closest to your fluorescence work) https://sigtuple.com/ · NVIDIA healthcare/MONAI + CUDA/TensorRT https://www.nvidia.com/en-us/about-nvidia/careers/ · GE HealthCare (Bengaluru R&D) https://www.gehealthcare.com/ · Microsoft Research India https://www.microsoft.com/en-us/research/lab/microsoft-research-india/ · Google Research India (stretch now, real by Tier 1) https://research.google/locations/india/

Tier 1 — broaden: global medical-imaging AI if open to remote/relocation — Aidoc, Rad AI, Annalise.ai, PathAI, Paige.AI, Owkin (Paris/NYC, imaging + TechBio) https://www.owkin.com/

Tier 2 — AI drug discovery + research labs (relocate): Recursion (post-Exscientia; *phenomics = deep learning on cell images — uniquely aligned with your medical-CV background applied to biology*) https://www.recursion.com/careers · Insitro https://insitro.com/ · Genesis Therapeutics https://www.genesistherapeutics.ai/ · Iambic https://www.iambic.ai/ · Chai Discovery https://www.chaidiscovery.com/ · EvolutionaryScale https://www.evolutionaryscale.ai/ · Profluent https://www.profluent.bio/ · InstaDeep (BioNTech; London/Paris) https://www.instadeep.com/ · Valence Labs (Recursion's research arm, Montreal; great RS-track practice) https://www.valencelabs.com/ · **Isomorphic Labs** https://www.isomorphiclabs.com/careers · **Google DeepMind** https://deepmind.google/about/careers/

- *Neutral nuance:* Isomorphic has a large drug-discovery partnership with Eli Lilly — not a blocker (people move between partnered orgs), but apply with your normal discretion.

Tier 3 — top research loops: Isomorphic, Google DeepMind, Google Research (global + India), Genentech gRED / Prescient Design, Microsoft Research Health Futures, Meta FAIR.

Conflict-of-interest guardrails (new in v6)

Not legal advice — verify specifics with the right people at Lilly.

1. **Public data only.** Use public benchmarks (Medical Decathlon/BraTS, QM9, PCQM4Mv2, CalMS21, PMC-OA, KITTI, etc.). Never Lilly proprietary data, targets, molecules, assays, or internal documents. In Capstone 3b use textbook molecules (H_2/LiH/H_2O/QM9), **not** your Lilly molecules (avoid the keto-enol / HATU work).
2. **Generic, published methods** in any portfolio artifact — no Lilly trade secrets or unpublished internal techniques.
3. **Personal time and resources** — your own hardware/accounts; not Lilly compute, licenses, or work hours.
4. **Clear before you publish** — run any paper/blog/repo/talk through Lilly's standard external-publication / IP-review process; when unsure, ask first.
5. **Check your agreement** — review employment / invention-assignment / moonlighting terms; confirm portfolio work is clear of them.
6. **Positioning stays generic** — your "why me" narrative is about transferable skills, never Lilly specifics.

Cross-cutting habits (every week, no exceptions)

1. **Paper-of-the-week** — one paper, deeply read, derivations redone on paper. Don't break the streak.
2. **Whiteboard Friday** — one concept whiteboarded without notes; photograph it. These become your interview-prep deck.
3. **Re-implementation Saturday** — one concept from scratch in <200 lines. Repo: `weekly-rebuild/wk01-linalg`, etc. *(This and Thread M reinforce each other.)*
4. **Sunday writeup** — a 1-page note explaining the week's deepest concept to a hypothetical junior. Teaching is the senior-level forcing function.
5. **Senior-engineer "war chest"** — one Obsidian/Notion doc collecting derivations, paper one-liners, gotchas, anti-patterns. Your interview safety net.
6. **Reproduce-a-target-paper (monthly)** — from Thread R; a target-team paper reproduced and written up publicly. *(New in v6.)*
7. **Positioning doc (living)** — the "why me for digital biology" narrative + two research job-talk decks alongside the war chest. *(New in v6.)*

Resource index (quick lookup)

Brush-up (P0) — StatQuest https://www.youtube.com/@statquest/playlists · https://statquest.org/ · 3B1B NN https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi

Foundations math (P1) — 3B1B LA https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab · 3B1B Calc https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr · MIT 18.06 https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/ · MIT 18.065 https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/ · Stat 110 https://projects.iq.harvard.edu/stat110/youtube · MIT RES.6-012 https://ocw.mit.edu/courses/res-6-012-introduction-to-probability-spring-2018/ · Matrix Cookbook https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf · Matrix Calculus https://explained.ai/matrix-calculus/ · Karpathy Z2H https://www.youtube.com/playlist?list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ · hub https://karpathy.ai/zero-to-hero.html

Later-threaded math — Boyd https://www.youtube.com/playlist?list=PL3940DD956CDF0622 · https://web.stanford.edu/~boyd/cvxbook/ · Ruder https://www.ruder.io/optimizing-gradient-descent/ · MacKay https://www.inference.org.uk/itila/book.html · Group theory https://www.youtube.com/results?search_query=group+theory+for+physicists+lectures · GDL lectures https://geometricdeeplearning.com/lectures/

Medical imaging (P2) — MRI Q&A https://mriquestions.com/index.html · OHBM https://www.youtube.com/@ohbmonline/playlists · Callaghan https://www.youtube.com/playlist?list=PLqEMVgX2VgZcG_xR1QRdsuc-X_zEgMx-Z · iBiology https://www.youtube.com/playlist?list=PLF513KEDjY9YBNhwMfX6jMl3PlpdW9TMP · AI for Medical Diagnosis https://www.coursera.org/learn/ai-for-medical-diagnosis · DigitalSreeni https://www.youtube.com/@DigitalSreeni/playlists · code https://github.com/bnsreenu/python_for_microscopists · Conv arithmetic https://arxiv.org/abs/1603.07285 · https://github.com/vdumoulin/conv_arithmetic · MONAI https://github.com/Project-MONAI/tutorials · https://www.youtube.com/@projectmonai/videos · nnU-Net https://github.com/MIC-DKFZ/nnUNet · TorchIO https://torchio.readthedocs.io/ · Stanford AIMI https://aimi.stanford.edu/education/educational-resources · CS231n https://www.youtube.com/playlist?list=PL3FW7Lu3i5JvHM8ljYj-zLfQRF3EO8sYv · https://cs231n.github.io/ · Metrics Reloaded https://www.nature.com/articles/s41592-023-02151-z · Seg papers: U-Net https://arxiv.org/abs/1505.04597 · V-Net https://arxiv.org/abs/1606.04797 · Attention U-Net https://arxiv.org/abs/1804.03999 · nnU-Net https://www.nature.com/articles/s41592-020-01008-z · Loss survey https://arxiv.org/abs/2006.14822 · Generalized Dice https://arxiv.org/abs/1707.03237 · Focal Tversky https://arxiv.org/abs/1810.07842 · Boundary loss https://arxiv.org/abs/1812.07032 · ViT https://arxiv.org/abs/2010.11929 · TransUNet https://arxiv.org/abs/2102.04306 · Swin-UNet https://arxiv.org/abs/2105.05537 · UNETR https://arxiv.org/abs/2103.10504 · Swin UNETR https://arxiv.org/abs/2201.01266 · SAM https://arxiv.org/abs/2304.02643 · MedSAM https://www.nature.com/articles/s41467-024-44824-z · MedNeXt https://arxiv.org/abs/2303.09975 · **Datasets:** Medical Decathlon http://medicaldecathlon.com/ · BraTS https://www.synapse.org/brats · TCIA https://www.cancerimagingarchive.net/ · IXI https://brain-development.org/ixi-dataset/ · LIVECell https://github.com/sartorius-research/LIVECell · Sartorius Kaggle https://www.kaggle.com/competitions/sartorius-cell-instance-segmentation · BBBC https://bbbc.broadinstitute.org/

Deep learning core (P3) — CS25 https://web.stanford.edu/class/cs25/ · https://www.youtube.com/playlist?list=PLoROMvodv4rNiJRchCzutFw5ItR_Z27CM · CS336 https://stanford-cs336.github.io/spring2024/ · https://www.youtube.com/playlist?list=PLoROMvodv4rOY23Y0BoGoBGgQ1zmU_MT_ · CS236 https://www.youtube.com/playlist?list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8 · Lilian Weng https://lilianweng.github.io/ · Annotated Transformer http://nlp.seas.harvard.edu/annotated-transformer/ · HF NLP https://huggingface.co/learn/nlp-course · HF CV https://huggingface.co/learn/computer-vision-course · HF Diffusion https://huggingface.co/learn/diffusion-course · RoPE https://arxiv.org/abs/2104.09864 · RMSNorm https://arxiv.org/abs/1910.07467 · FlashAttention https://arxiv.org/abs/2205.14135 · GQA https://arxiv.org/abs/2305.13245 · SwiGLU https://arxiv.org/abs/2002.05202 · Switch/MoE https://arxiv.org/abs/2101.03961 · ConvNeXt https://arxiv.org/abs/2201.03545 · DINOv2 https://arxiv.org/abs/2304.07193 · MAE https://arxiv.org/abs/2111.06377 · DDPM https://arxiv.org/abs/2006.11239 · Score-based https://yang-song.net/blog/2021/score/ · Diffusion (blog) https://lilianweng.github.io/posts/2021-07-11-diffusion-models/

Quantum (P4) — IBM Quantum Learning https://learning.quantum.ibm.com/ · Watrous https://www.youtube.com/playlist?list=PLOFEBzvs-VvrXTMy5Y2IqmSaUjfnhvBHR · Basics https://learning.quantum.ibm.com/course/basics-of-quantum-information · Algorithms https://learning.quantum.ibm.com/course/fundamentals-of-quantum-algorithms · General Formulation https://learning.quantum.ibm.com/course/general-formulation-of-quantum-information · Error Correction https://learning.quantum.ibm.com/course/foundations-of-quantum-error-correction · Qiskit https://www.youtube.com/@qiskit/playlists · QDA blog https://www.ibm.com/quantum/blog/iql-migration · PennyLane codebook https://pennylane.ai/codebook · PennyLane QML https://pennylane.ai/topics/quantum-machine-learning · PennyLane chemistry https://pennylane.ai/qml/demos_quantum-chemistry · MIT 8.04 https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/ · Quantum Country https://quantum.country/qcvc · Schuller https://www.youtube.com/playlist?list=PLPH7f_7ZlzxTi6kS4vCmv4ZKm9u8g5yic · Péré https://github.com/Christophe-pere/QML-Course · Hjorth-Jensen https://github.com/CompPhysics/QuantumComputingMachineLearning · VQE https://www.nature.com/articles/ncomms5213 · ADAPT-VQE https://www.nature.com/articles/s41467-019-10988-2 · DMET-VQE https://arxiv.org/abs/2108.08987 · QAOA https://arxiv.org/abs/1411.4028 · Hadfield https://arxiv.org/abs/1709.03489 · QML=kernels https://arxiv.org/abs/2101.11020 · Barren plateaus https://www.nature.com/articles/s41467-018-07090-4 · Expressibility https://arxiv.org/abs/1905.10876 · Advantage-goal https://arxiv.org/abs/2203.01340 · PQC encodings https://arxiv.org/abs/2008.08605 · Pauli grouping https://arxiv.org/abs/1907.13117

Production / MLOps (P5) — CS329S https://stanford-cs329s.github.io/ · Made With ML https://madewithml.com/ · Full Stack DL https://fullstackdeeplearning.com/course/2022/ · Chip Huyen https://huyenchip.com/ml-interviews-book/ · Hamel Husain https://www.oreilly.com/radar/what-we-learned-from-a-year-of-building-with-llms-part-i/ · Eugene Yan patterns https://eugeneyan.com/writing/llm-patterns/ · Eugene Yan evals https://eugeneyan.com/writing/llm-evaluators/ · Anthropic contextual retrieval https://www.anthropic.com/news/contextual-retrieval · Anthropic Constitutional AI https://www.anthropic.com/news/constitutional-ai · LlamaIndex https://docs.llamaindex.ai/ · RAGAS https://docs.ragas.io/ · DSPy https://dspy.ai/ · Raschka https://magazine.sebastianraschka.com/ · HF multi-GPU https://huggingface.co/docs/transformers/en/perf_train_gpu_many · vLLM https://arxiv.org/abs/2309.06180 · ZeRO https://arxiv.org/abs/1910.02054

DSA & C++ (Thread T) — learncpp https://www.learncpp.com/ · The Cherno https://www.youtube.com/@TheCherno · cppreference https://en.cp preference.com/ · NeetCode https://neetcode.io/ · LeetCode https://leetcode.com/ · MIT 6.006 https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/ · Abdul Bari https://www.youtube.com/@abdul_bari

Implement-by-hand ML (Thread M) — new — d2l.ai https://d2l.ai/ · labml.ai https://nn.labml.ai/ · Annotated Transformer http://nlp.seas.harvard.edu/annotated-transformer/ · micrograd https://github.com/karpathy/micrograd · nanoGPT https://github.com/karpathy/nanoGPT · DeepMind RE round guide https://igotanoffer.com/en/advice/google-deepmind-research-engineer-interview

Research output (Thread R) — new — Papers with Code https://paperswithcode.com/ · MICCAI https://www.miccai.org/ · MIDL https://2025.midl.io/ · MLSB https://www.mlsb.io/ · LoG https://logconference.org/ · ISCB/ISMB https://www.iscb.org/ · arXiv https://arxiv.org/ · bioRxiv https://www.biorxiv.org/ · OpenReview https://openreview.net/

GenAI & Agents (P7) — HF LLM course https://huggingface.co/learn/llm-course · HF Agents https://huggingface.co/learn/agents-course · HF MCP https://huggingface.co/learn/mcp-course · Anthropic Building Effective Agents https://www.anthropic.com/research/building-effective-agents · DeepLearning.AI short courses https://www.deeplearning.ai/short-courses/ · LangGraph https://langchain-ai.github.io/langgraph/ · LoRA https://arxiv.org/abs/2106.09685 · QLoRA https://arxiv.org/abs/2305.14314 · InstructGPT/RLHF https://arxiv.org/abs/2203.02155 · DPO https://arxiv.org/abs/2305.18290 · CLIP https://arxiv.org/abs/2103.00020 · LLaVA https://arxiv.org/abs/2304.08485 · Flamingo https://arxiv.org/abs/2204.14198 · **Corpora:** PMC-OA https://www.ncbi.nlm.nih.gov/pmc/tools/openftlist/ · PubMed E-utilities https://www.ncbi.nlm.nih.gov/books/NBK25501/ · Semantic Scholar API https://www.semanticscholar.org/product/api

Computer Vision expert + Video (P8) — First Principles of CV https://fpcv.cs.columbia.edu/ · FPCV Coursera https://www.coursera.org/specializations/firstprinciplesofcomputervision · Cyrill Stachniss https://www.youtube.com/@CyrillStachniss · Michigan EECS 498 https://web.eecs.umich.edu/~justincj/teaching/eecs498/ · playlist https://www.youtube.com/playlist?list=PL5-TkQAfAZFbzxjBHtzdVCWE0Zbhomg7r · NeRF https://arxiv.org/abs/2003.08934 · 3D Gaussian Splatting https://arxiv.org/abs/2308.04079 · OpenCV https://docs.opencv.org/ · PyImageSearch https://pyimagesearch.com/ · ONNX https://onnx.ai/ · TensorRT https://developer.nvidia.com/tensorrt · **Video (new):** VideoMAE https://arxiv.org/abs/2203.12602 · TimeSformer https://arxiv.org/abs/2102.05095 · ViViT https://arxiv.org/abs/2103.15691 · SlowFast https://arxiv.org/abs/1812.03982 · SAM 2 https://github.com/facebookresearch/sam2 · RAFT https://github.com/princeton-vl/RAFT · ByteTrack https://github.com/ifzhang/ByteTrack · OC-SORT https://github.com/noahcao/OC_SORT · BoT-SORT https://github.com/NirAharon/BoT-SORT · HOTA https://arxiv.org/abs/2009.07736 · MOTChallenge https://motchallenge.net/ · DAVIS https://davischallenge.org/ · YouTube-VOS https://youtube-vos.org/ · **Life-sciences video (new):** DeepLabCut https://www.deeplabcut.org/ · SLEAP https://sleap.ai/ · Cellpose https://www.cellpose.org/ · TrackMate https://imagej.net/plugins/trackmate/ · Ultrack https://github.com/royerlab/ultrack · Cell Tracking Challenge http://celltrackingchallenge.net/ · CalMS21 https://data.caltech.edu/records/s0vdx-0k302 · **Deploy datasets:** KITTI https://www.cvlibs.net/datasets/kitti/ · TUM RGB-D https://cvg.cit.tum.de/data/datasets/rgbd-dataset · ETH3D https://www.eth3d.net/

Geometric / Molecular ML (P6) — deepened — CS224W https://web.stanford.edu/class/cs224w/ · https://snap.stanford.edu/class/cs224w-2023/ · playlist https://www.youtube.com/playlist?list=PLoROMvodv4rPLKxIpqhjhPgdQy7imNkDn · Bronstein GDL https://geometricdeeplearning.com/lectures/ · proto-book https://arxiv.org/abs/2104.13478 · PyG https://pytorch-geometric.readthedocs.io/ · DGL https://www.dgl.ai/ · DeepChem https://deepchem.io/tutorials/ · TDC https://tdcommons.ai/ · SchNet https://arxiv.org/abs/1706.08566 · DimeNet++ https://arxiv.org/abs/2011.14115 · GemNet https://arxiv.org/abs/2106.08903 · MolFormer https://arxiv.org/abs/2106.09553 · Equivariant (Cohen&Welling) https://arxiv.org/abs/1602.07576 · EGNN https://arxiv.org/abs/2102.09844 · e3nn https://e3nn.org/ · https://github.com/e3nn/e3nn · e3nn tutorial https://blondegeek.github.io/e3nn_tutorial/ · NequIP https://github.com/mir-group/nequip · https://www.nature.com/articles/s41467-022-29939-5 · MACE https://github.com/ACEsuit/mace · https://arxiv.org/abs/2206.07697 · Allegro https://github.com/mir-group/allegro · Equiformer https://arxiv.org/abs/2206.11990 · curated list https://github.com/Eipgen/Neural-Network-Models-for-Chemistry · AlphaFold 2 https://www.nature.com/articles/s41586-021-03819-2 · ESM https://github.com/facebookresearch/esm · ESM Atlas https://esmatlas.com/ · Boltz https://github.com/jwohlwend/boltz · OpenFold https://github.com/aqlaboratory/openfold · **Datasets:** QM9 https://quantum-machine.org/datasets/ · PCQM4Mv2 https://ogb.stanford.edu/kddcup2021/pcqm4m/ · MoleculeNet https://moleculenet.org/ · OGB https://ogb.stanford.edu/

Big-tech interview (P9) — NeetCode https://neetcode.io/ · LeetCode https://leetcode.com/ · DesignGurus https://www.designgurus.io/ · ByteByteGo https://bytebytego.com/ · https://www.youtube.com/@ByteByteGo · Gaurav Sen https://www.youtube.com/@gkcs · MIT 6.824 https://pdos.csail.mit.edu/6.824/

Company ladder (new) — Qure.ai https://qure.ai/ · SigTuple https://sigtuple.com/ · NVIDIA https://www.nvidia.com/en-us/about-nvidia/careers/ · GE HealthCare https://www.gehealthcare.com/ · MSR India https://www.microsoft.com/en-us/research/lab/microsoft-research-india/ · Google Research India https://research.google/locations/india/ · Owkin https://www.owkin.com/ · Recursion https://www.recursion.com/careers · Insitro https://insitro.com/ · Genesis https://www.genesistherapeutics.ai/ · Iambic https://www.iambic.ai/ · Chai https://www.chaidiscovery.com/ · EvolutionaryScale https://www.evolutionaryscale.ai/ · Profluent https://www.profluent.bio/ · InstaDeep https://www.instadeep.com/ · Valence Labs https://www.valencelabs.com/ · Isomorphic Labs https://www.isomorphiclabs.com/careers · Google DeepMind https://deepmind.google/about/careers/

A note on what you're taking on

This is a ~16.5-month, ~1,950-hour program on top of a full-time senior role — genuinely ambitious, and only sustainable as a marathon. Anchors: the self-tests and capstones are the real gates, not the calendar dates, which are planning aids you can flex; the three threads are your shock absorbers — dial them to ~2–3 hrs/week combined whenever a week gets heavy, since consistency beats volume for muscle memory; protect one true rest block a week plus the buffer weeks, because burnout in month five costs more than any schedule it appears to save. Start applying at Tier 0 and keep learning through your loops — the plan makes you better mid-interview, so don't pause it. Re-activating rusty fundamentals and re-deriving what you already use will feel like going backwards at first — it isn't; it's exactly what turns "I use this" into "I can teach this to a panel." Adjust as you go: it's a scaffold for durable mastery and a real shot at a life-sciences research seat, not a race to the bottom of your energy.